UDAAN COACHING CENTRE

CLASS X • MATHEMATICS

40 MOST IMPORTANT

BOARD PREVIOUS-YEAR QUESTIONS

CHAPTER 10 - CIRCLES

Medium	English
Class	10 (X)
Board tags in source	2023 • 2024 • 2025
Selection	Only questions explicitly labelled BOARD in the supplied question bank
Practice questions	Not included

Diagram accuracy note
Where the supplied source contains a figure, the figure below the question is taken from that same source question. No substitute or unrelated diagrams have been inserted. Text has been cleaned for obvious spelling, punctuation and unit mismatches while mathematical values and board-year tags are retained.

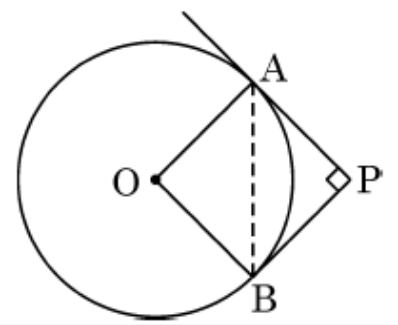
SECTION A | 1-MARK BOARD PYQs

Choose the correct option where options are provided.

Q1. The tangents drawn at the extremities of the diameter of a circle are always: [BOARD 2025]

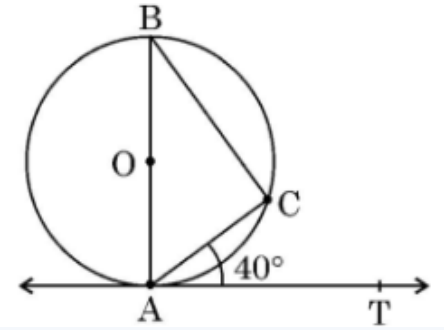
(a) parallel
(b) perpendicular
(c) equal
(d) intersecting

Q2. In the given figure, tangents PA and PB to the circle with centre O, from point P, are perpendicular to each other. If PA = 5 cm, then the length of AB is: [BOARD 2024]



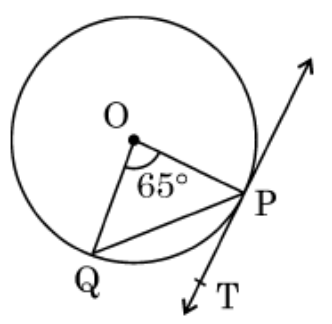
(a) 5 cm
(b) $5\sqrt{2}$ cm
(c) $2\sqrt{5}$ cm
(d) 10 cm

Q3. In the given figure, AT is tangent to a circle with centre O. If $\angle CAT = 40°$, then $\angle CBA$ is equal to: [BOARD 2024 & 2025]



(a) 70°
(b) 50°
(c) 65°
(d) 40°

Q4. In the given figure, PT is tangent to a circle with centre O. Chord PQ subtends an angle of 65° at the centre. The measure of ∠QPT is: [BOARD 2023 & 2024]

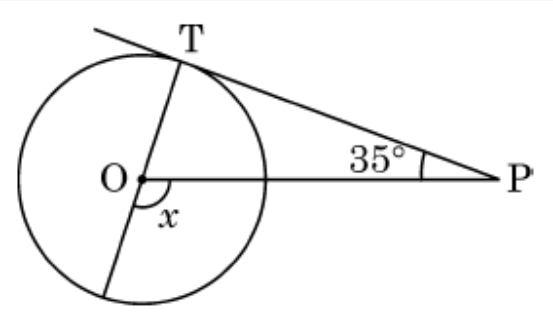


(a) 65°
(b) 57.5°
(c) 67.5°
(d) 32.5°

Q5. The maximum number of common tangents that can be drawn to two circles intersecting at two distinct points is: [BOARD 2024]

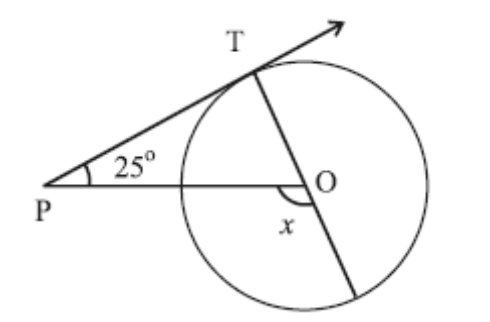
(a) 4
(b) 3
(c) 2
(d) 1

Q6. In the given figure, PT is a tangent to a circle with centre O. If ∠TPO = 35°, then the measure of ∠x is: [BOARD 2024 & 2025]



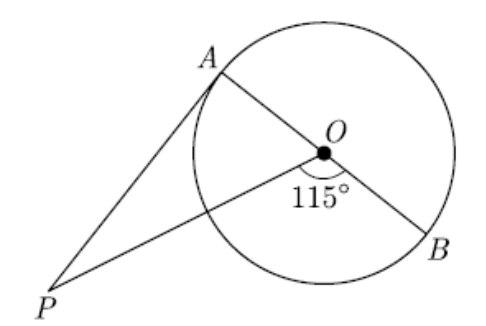
(a) 110°
(b) 115°
(c) 120°
(d) 125°

Q7. In the given figure, PT is tangent at T to the circle with centre O. If ∠TPO = 25°, then x is: [BOARD 2023]

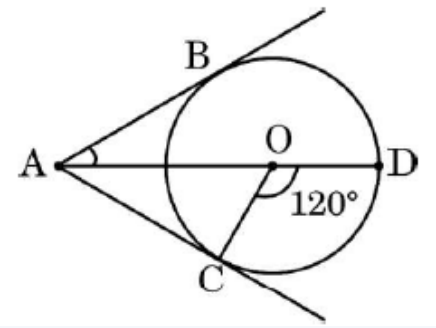


(a) 25°
(b) 65°
(c) 90°
(d) 115°

Q8. In the given figure, PA is a tangent from an external point P to a circle with centre O. If ∠POB = 115°, then find ∠APO. [BOARD 2025]

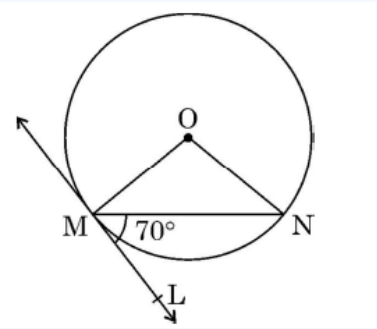


Q9. In the given figure, AC and AB are tangents to a circle with centre O. If ∠COD = 120°, then ∠BAO is: [BOARD 2023]



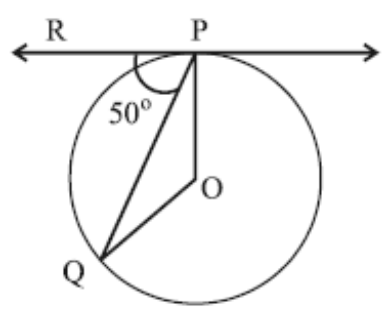
(a) 30°
(b) 60°
(c) 45°
(d) 90°

Q10. In the given figure, O is the centre of the circle. MN is a chord, and the tangent ML at M makes an angle of 70° with MN. The measure of ∠MON is: [BOARD 2024]



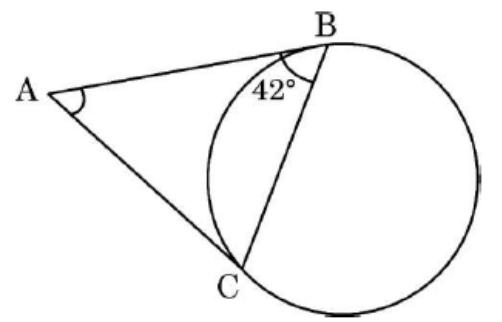
(a) 120°
(b) 140°
(c) 70°
(d) 90°

Q11. In the given figure, O is the centre of the circle and PQ is a chord. If the tangent PR at P makes an angle of 50° with PQ, then ∠POQ is: [BOARD 2023]



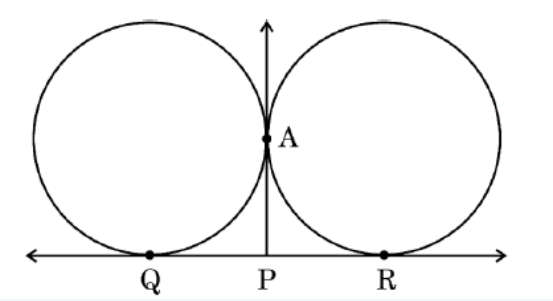
(a) 50°
(b) 40°
(c) 100°
(d) 130°

Q12. In the given figure, AB and AC are tangents to the circle. If ∠ABC = 42°, then the measure of ∠BAC is: [BOARD 2024]



(a) 96°
(b) 42°
(c) 106°
(d) 86°

Q13. In the given figure, QR is a common tangent to two circles touching externally at A. The tangent at A meets QR at P. If AP = 4.2 cm, find the length of QR. [BOARD 2024]

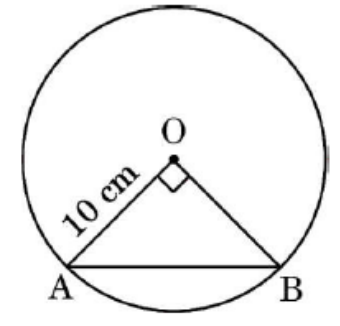


(a) 4.2 cm
(b) 2.1 cm
(c) 8.4 cm
(d) 6.3 cm

Q14. The area of a square inscribed in a circle of radius $5\sqrt{2}$ cm is: [BOARD 2024]

(a) 50 cm²
(b) 100 cm²
(c) 25 cm²
(d) 200 cm²

Q15. A chord of a circle of radius 10 cm subtends a right angle at its centre. The length of the chord (in cm) is: [BOARD 2024]

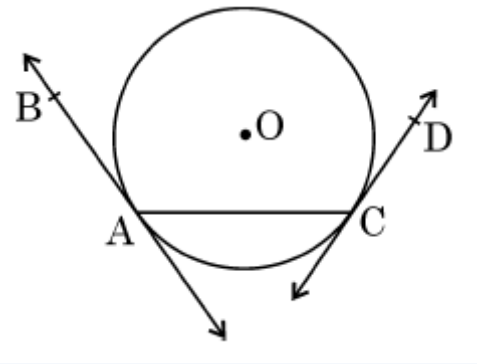


(a) $5\sqrt{2}$
(b) $10\sqrt{2}$
(c) $5/\sqrt{2}$
(d) 5

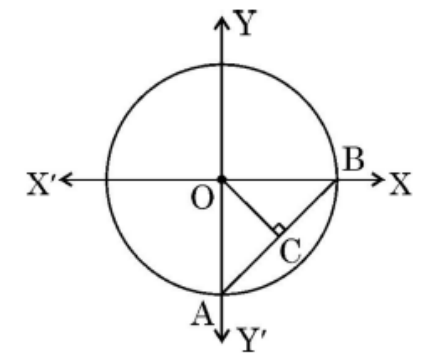
SECTION B | 2-MARK BOARD PYQs

Write the required justification or calculation steps.

Q16. In the given figure, AB and CD are tangents to a circle with centre O. Is $\angle BAC = \angle DCA$? Justify your answer. [BOARD 2024]

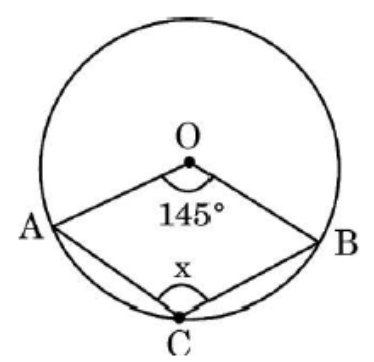


Q17. In the given figure, a circle with centre O has radius 7 cm, and OC is a median of ΔOAB. Find the length of the median OC. [BOARD 2024]

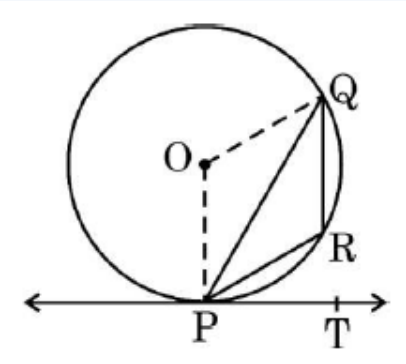


Q18. From an external point P, two tangents PA and PB are drawn to a circle with centre O. Prove that OP is the perpendicular bisector of chord AB. [BOARD 2024]

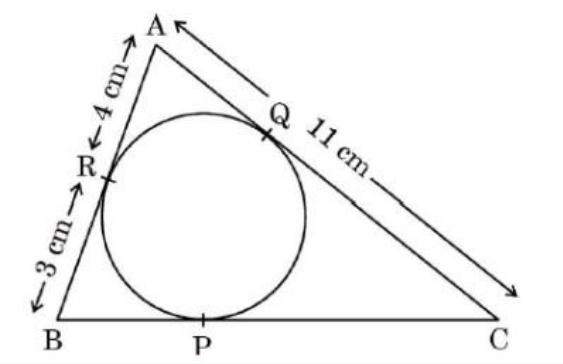
Q19. In the given figure, O is the centre of the circle. If ∠AOB = 145°, find the value of x. [BOARD 2024]



Q20. In the given figure, PQ is a chord of the circle with centre O, and PT is a tangent at P. If ∠QPT = 55°, find ∠PRQ. [BOARD 2023]



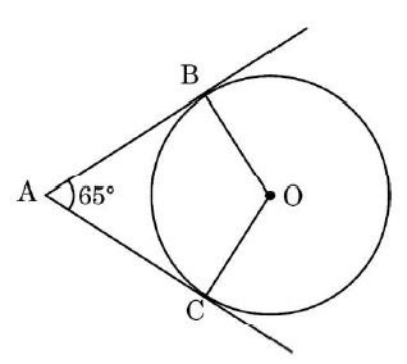
Q21. In the given figure, ΔABC circumscribes a circle. Find the length of BC if AR = 4 cm, BR = 3 cm and AC = 11 cm. [BOARD 2024]



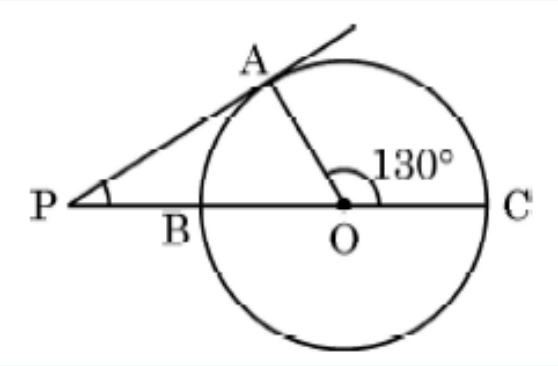
Q22. If two tangents inclined at an angle of 60° are drawn to a circle of radius 3 cm, find the length of each tangent. [BOARD 2024]

Q23. Prove that the tangents drawn at the ends of a diameter of a circle are parallel. [BOARD 2023, 2024 & 2025]

Q24. In the given figure, O is the centre of the circle. AB and AC are tangents drawn from point A. If ∠BAC = 65°, find the measure of ∠BOC. [BOARD 2023]



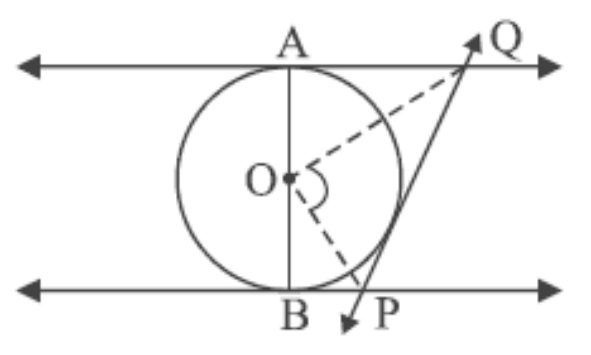
Q25. In the given figure, PA is a tangent to the circle from an external point P, and PBC is a secant to the circle with BC as diameter. If ∠AOC = 130°, find ∠APB. [BOARD 2023]



SECTION C | 3-MARK BOARD PYQs

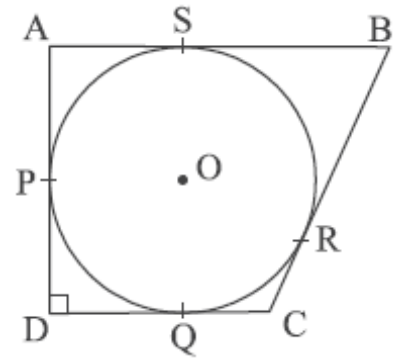
Show the main steps of the proof or calculation.

Q26. In the given figure, AB is a diameter of the circle with centre O. AQ, BP and PQ are tangents to the circle. Prove that ∠POQ = 90°. [BOARD 2024 & 2025]

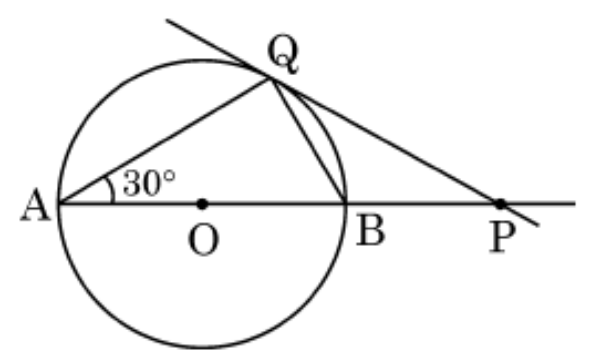


Q27. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. [BOARD 2025]

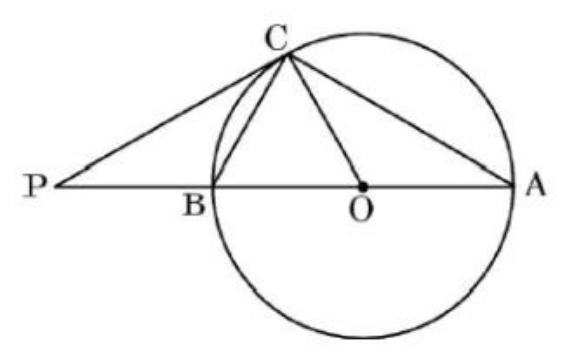
Q28. A circle with centre O and radius 8 cm is inscribed in quadrilateral ABCD, where P, Q, R and S are the points of contact shown. If AD ⊥ DC, BC = 30 cm and BS = 24 cm, find DC. [BOARD 2024]



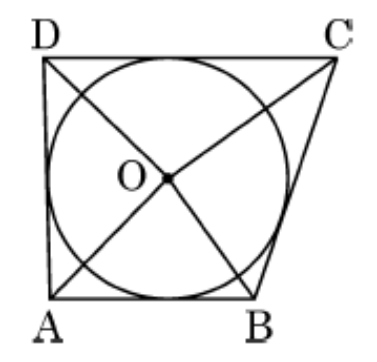
Q29. In the given figure, PQ is tangent to a circle with centre O and ∠BAQ = 30°. Show that BP = BQ. [BOARD 2024]



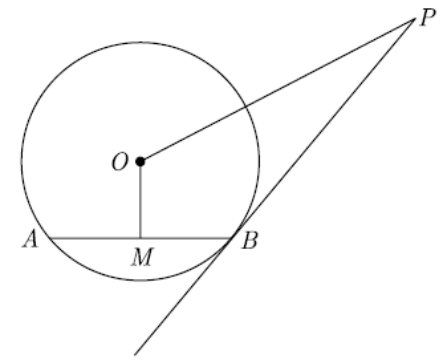
Q30. In the given figure, PC is tangent to a circle at C. Diameter AOB, when extended, meets the tangent at P. Find ∠CBA and ∠BCO if ∠PCA = 110°. [BOARD 2025]



Q31. In the given figure, AB, BC, CD and DA are tangents to the circle with centre O, forming quadrilateral ABCD. Show that ∠AOB + ∠COD = 180°. [BOARD 2024]

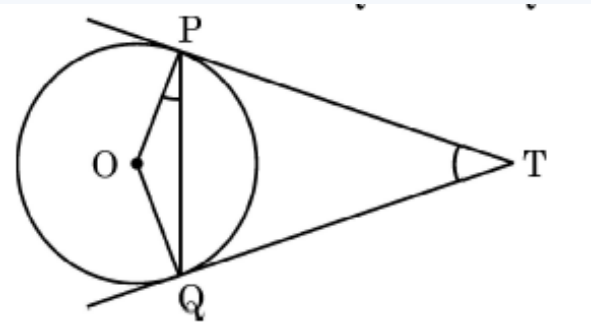


Q32. PB is a tangent to a circle with centre O at B. AB is a chord 24 cm long and is at a distance of 5 cm from the centre. If PB = 20 cm, find PO. [BOARD 2025]

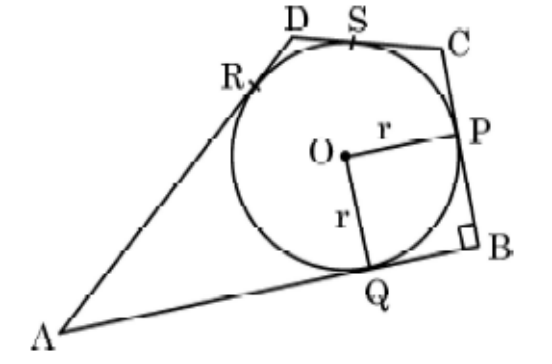


Q33. Prove that a parallelogram circumscribing a circle is a rhombus. [BOARD 2024 & 2025]

Q34. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that ∠PTQ = 2∠OPQ. [BOARD 2023]



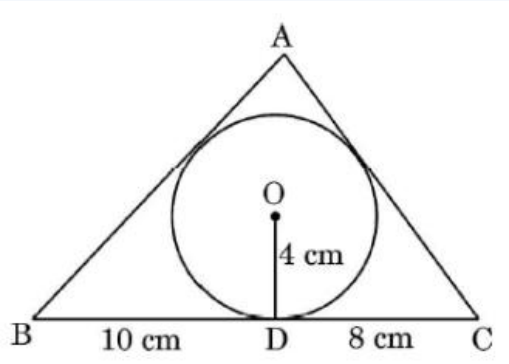
Q35. In the given figure, a circle is inscribed in quadrilateral ABCD, where ∠B = 90°. If AD = 17 cm, AB = 20 cm and DS = 3 cm, find the radius of the circle. [BOARD 2023]



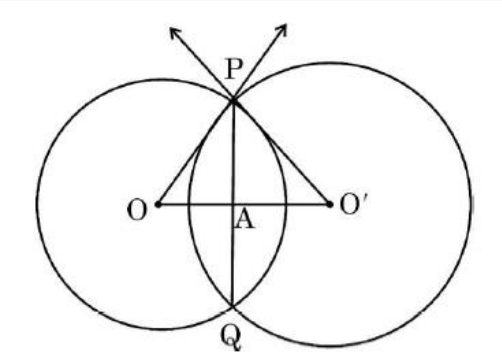
SECTION D | 5-MARK BOARD PYQs

Write a complete proof or solution, including the required reasoning.

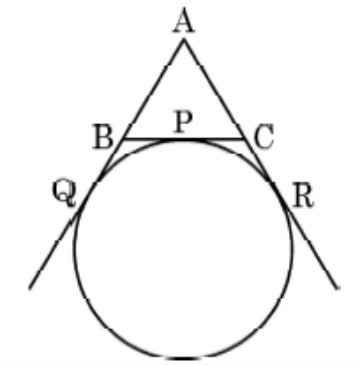
Q36. A triangle ABC circumscribes a circle of radius 4 cm. The segments BD and DC are 10 cm and 8 cm, respectively. Find AB and AC if the area of ΔABC is 90 cm². [BOARD 2023]



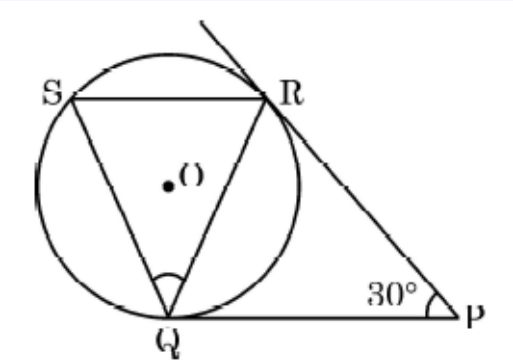
Q37. Two circles with centres O and O′ and radii 6 cm and 8 cm, respectively, intersect at P and Q. If OP and O′P are perpendicular to each other, find the length of the common chord PQ. [BOARD 2023]



Q38. A circle touches side BC of ΔABC at P and touches AB and AC, when produced, at Q and R respectively. Show that AQ = ½ × (perimeter of ΔABC). [BOARD 2023]



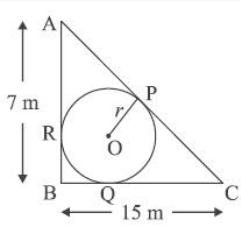
Q39. In the given figure, tangents PQ and PR are drawn to a circle such that ∠RPQ = 30°. A chord RS is drawn parallel to tangent PQ. Find ∠RQS. [BOARD 2023]

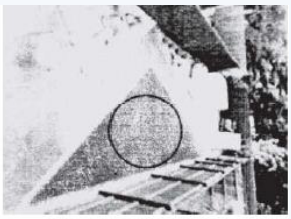


SECTION E | CASE-BASED BOARD PYQ

Read the case and answer each part.

Q40. A backyard is in the shape of a triangle ABC, right-angled at B. AB = 7 m and BC = 15 m. A circular pit is dug inside it so that the circle touches AC, BC and AB at P, Q and R, respectively, and AP = x m. [BOARD 2024]





(i) Find the length of AR in terms of x. (1 mark)

(ii) (a) Find PC in terms of x and hence find x. (2 marks)

OR

(ii) (b) Find x and hence find the radius r of the circle. (2 marks)

(iii) State the type of quadrilateral BQOR. (1 mark)

FINAL PAGE | INSTRUCTIONS FOR STUDENTS

How to attempt this worksheet

- 1. Attempt all 40 questions. The worksheet contains only questions carrying a BOARD year tag in the supplied question bank; no untagged practice questions have been added.
- 2. Read the figure labels carefully. For questions with a source figure, use the exact labels, points, angles and values shown in the diagram immediately below that question.
- 3. Draw diagrams neatly in your answer sheet. Use a ruler and compass where suitable, label every point, and mark the given angles and lengths clearly.
- 4. Show complete working. For 2-mark, 3-mark and 5-mark questions, write the theorem or property used and show the main calculation/proof steps. Do not write only the final answer.
- 5. Use correct mathematical notation. Write cm or m as given, include square units for area, and distinguish radius, diameter, chord, tangent and secant.
- 6. For proof questions, state the reason for each important equality or angle relation, especially the tangent-radius theorem and the equal-tangents theorem.
- 7. For the case-based question, answer each subpart separately. Choose only one of the alternatives where an OR is provided.
- 8. No calculator is required. Keep your answers clear, well-spaced and easy to check.

UDAAN COACHING CENTRE
Prepared for Class X Mathematics - Chapter 10: Circles
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